

Waqar Hassan

Postdoctoral Researcher | Graph Machine Learning | Spatio-temporal AI | Explainable AI

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Professional Profile

Postdoctoral researcher developing graph-based machine learning methods for modeling, interpreting, and visualizing complex spatio-temporal systems. My research combines Graph Neural Networks, representation learning, explainable AI, and visual analytics to develop interpretable machine learning methods for complex urban systems and real-world decision support. I have published in international peer-reviewed conferences including IEEE ICDM, IJCAI, IEEE DSAA, BRACIS, and SIBGRAPI, and enjoy building scalable, interpretable AI systems with practical impact.

Research Expertise

- Graph Machine Learning
- Graph Neural Networks
- Explainable AI
- Heterogeneous Data Integration
- Spatio-temporal Learning
- Representation Learning
- Visual Analytics
- Urban Data Science

Professional Experience

Postdoctoral Researcher

Feb. 2024–Present

University of São Paulo — São Carlos, Brazil

- Developed Graph Neural Network models for spatio-temporal prediction over large-scale urban street networks.
- Designed a scalable graph-construction framework integrating street topology, socioeconomic variables, and temporally evolving event data into heterogeneous graph representations.
- Proposed multimodal representation-learning methods using Graph Autoencoders to combine static and dynamic urban information.
- Designed explainable-AI techniques based on dimensionality reduction, inverse projection, and latent-space analysis to interpret graph embeddings and model behavior.
- Built interactive visual analytics tools for exploring graph embeddings, feature relationships, and spatio-temporal prediction results.
- Implemented reusable software frameworks for graph-based machine learning research and experimentation across multiple urban prediction tasks.

Software Engineer II

Dec. 2021–Mar. 2023

LexisNexis Risk Solutions — São Paulo, Brazil

- Developed distributed ETL pipelines using HPCC Systems, processing more than 100 million records per day and improving data throughput by approximately 30%.
- Automated data-cleaning and parsing workflows, reducing manual processing effort by approximately 40%.
- Integrated machine-learning models into ETL workflows, improving trend-detection accuracy by up to 25%.

Graduate Researcher

Mar. 2018–Jul. 2022

University of São Paulo — São Carlos, Brazil

- Investigated quantification and classification methods for dengue mosquito prediction using sensor-equipped mosquito traps.
- Designed and implemented a lightweight, memory-efficient quantification framework for high-speed data streams and large-scale data environments.
- Proposed a fast and accurate quantification method with applications in public-health monitoring; the work received the Best Paper Award at IEEE DSAA 2020.
- Developed deployment-oriented methods with low memory and computational requirements to support scalable field use.

Education

Ph.D. in Computer Science

Mar. 2018–Jul. 2022

University of São Paulo — Brazil

Research area: Data Science

M.Sc. in Electronics Engineering

Feb. 2014–Sep. 2016

International Islamic University Islamabad — Pakistan

Specialization: Machine Learning

CGPA: 3.90/4.00

Publications

- 2025** **Exploring Urban Factors with Autoencoders: Relationship Between Static and Dynamic Features**
Conference on Graphics, Patterns and Images (SIBGRAPI)
DOI: [10.1109/SIBGRAPI67909.2025.11223351](https://doi.org/10.1109/SIBGRAPI67909.2025.11223351)
- 2024** **Modeling and Predicting Crimes in the City of São Paulo Using Graph Neural Networks**
Brazilian Conference on Intelligent Systems (BRACIS)
DOI: [10.1007/978-3-031-79035-5_26](https://doi.org/10.1007/978-3-031-79035-5_26)
- 2021** **Accurately Quantifying under Score Variability**
IEEE International Conference on Data Mining (ICDM)
DOI: [10.1109/ICDM51629.2021.00149](https://doi.org/10.1109/ICDM51629.2021.00149)
- 2021** **Pitfalls in Quantification Assessment**
First International Workshop on Learning to Quantify: Methods and Applications, CIKM
CEUR Workshop Proceedings
- 2020** **Accurately Quantifying a Billion Instances per Second**
IEEE International Conference on Data Science and Advanced Analytics (DSAA)
Best Paper Award. DOI: [10.1109/DSAA49011.2020.00012](https://doi.org/10.1109/DSAA49011.2020.00012)
- 2020** **The Importance of the Test Set Size in Quantification Assessment**
International Joint Conference on Artificial Intelligence (IJCAI)
DOI: [10.24963/ijcai.2020/366](https://doi.org/10.24963/ijcai.2020/366)
- Preprint** **Enriching Dimensionality Reduction with Distortion Cues**
Journal manuscript (under review), available as SSRN preprint
DOI: [10.2139/ssrn.6604227](https://doi.org/10.2139/ssrn.6604227)

Technical Expertise

Graph Learning	Graph Neural Networks, Graph Autoencoders, representation learning, multimodal learning
Machine Learning	Classification, regression, clustering, statistical learning, quantification, ensemble Learning, cross-validation, hyperparameter optimization
Spatio-temporal Modeling	Graph-based prediction, urban network analysis, time-series forecasting, temporal modeling, ARIMA
Explainable AI and Visual Analytics	Dimensionality reduction, inverse projection, latent representation analysis, interactive visualization, Matplotlib, Plotly, Tableau
Programming	Python, PyTorch, PyTorch Geometric, TensorFlow, Scikit-learn, Pandas, NumPy, SQL, ECL
Scientific Computing & Development Tools	Git, Poetry, MLflow, Jupyter Notebook, VS Code, Overleaf, Jira, HPC Systems, Apache Spark

Awards and Fellowships

FAPESP Postdoctoral Fellowship — São Paulo Research Foundation	Postdoctoral
TWAS–CNPq Doctoral Fellowship — TWAS / CNPq	Ph.D.
Best Paper Award — IEEE DSAA	2020

Academic Service

- Mentored one Ph.D. student and one M.Sc. student in graph machine learning research.

Languages

English	Fluent
Portuguese	Intermediate
Urdu	Native